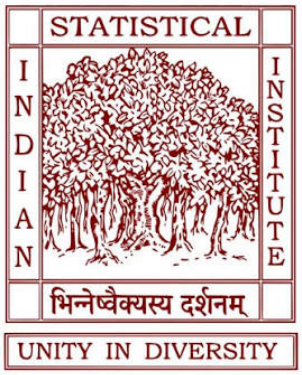
Linear Algebra : Assignment 3

Assignment submitted to

Indian Statistical Institute

by

Anik Barury CS 2604



DEPARTMENT OF COMPUTER SCIENCE
INDIAN STATISTICAL INSTITUTE
203 BARRACKPORE TRUNK ROAD, KOLKATA - 700108
WEST BENGAL, INDIA

September, 2026

1 Problem 1

(a) The operator T satisfies the equation

$$(T - I)(T - 2I)(T^2 + I) = 0.$$

The polynomials $p_1(x) = x - 1$, $p_2(x) = x - 2$, and $p_3(x) = x^2 + 1$ are pairwise co-prime. Since their product annihilates T, the Primary Decomposition Theorem guarantees that V decomposes as the direct sum of the kernels of these operators. So

$$V = \ker(T - I) \oplus \ker(T - 2I) \oplus \ker(T^2 + I) = U_1 \oplus U_2 \oplus U_3.$$

(b) Let $v \in U_3$. Then $(T^2 + I)v = 0$. We must show $Tv \in U_3$, i.e. that $(T^2 + I)(Tv) = 0$. Since polynomials evaluated at the same operator $p_1(x) = x - 1$– commute, we have

$$(T^2 + I)p_1(x) = x - 1 - -(T^2 + I)p_1(x) = x - 1 - -(0) = 0.$$

Thus $p_1(x) = x - 1$– and $p_1(x) = x - 1$– is $p_1(x) = x - 1$—invariant.

(c) On the subspace $p_1(x) = x - 1$–, we have $p_1(x) = x - 1$–. The minimal polynomial of $p_1(x) = x - 1$– must divide $p_1(x) = x-1$–. Since $p_2(x) = x-2$– is a real vector space, this polynomial has no real roots. Thus $p_2(x) = x - 2$– has no real eigenvalues and its characteristic polynomial must be a power of $p_2(x) = x - 2$–. Since the degree of the characteristic polynomial equals the dimension of the space, $p_2(x) = x - 2$– must be an even integer (or 0). Thus,

$$\dim(U_3) \in \{0, 2, 4, 6, \dots\}.$$

(d) Given $p_2(x) = x - 2$– and that $p_2(x) = x - 2$– has no eigenvalue 1. Since the absence of eigenvalue 1 implies

$$\ker(T - I) = U_1 = \{0\},$$

then $p_2(x) = x - 2$– and

$$\dim(U_2) + \dim(U_3) = 7.$$

From (c), $p_2(x) = x-2$– must be an even non-negative integer and so its possible values are $0, 2, 4$ or 6. The corresponding dimensions for $p_2(x) = x-2$– are $7, 5, 3$ or 1. The possible dimension pairs $p_2(x) = x-2$– are

$$(7, 0),\ (5, 2),\ (3, 4),\ (1, 6).$$

2 Problem 2

As stated, the problem is inconsistent. If $p_3(x) = x^2 + 1$– is a real vector space of odd dimension and $p_3(x) = x^2 + 1$–, then the characteristic polynomial of $p_3(x) = x^2 + 1$– has odd degree and therefore at least one real root. Hence $p_3(x) = x^2 + 1$– must have a real eigenvalue, contradicting the hypothesis that $p_3(x) = x^2 + 1$– has no eigenvalues. To make the statement meaningful, we assume instead that $p_3(x) = x^2 + 1$– is a real vector space (of even dimension) and that $p_3(x) = x^2 + 1$– has no real eigenvalues. Under this corrected hypothesis, the derivation proceeds as follows. Since $p_3(x) = x^2 + 1$– is a real vector space, the minimal polynomial of $p_3(x) = x^2 + 1$– factors into irreducible linear and quadratic terms. Since $p_3(x) = x^2 + 1$– has no real eigenvalues, it has no linear factors. Thus, there must exist an irreducible quadratic factor T– dividing the minimal polynomial, with T–. Consequently, T–. Take T–. Let T–. T– is T—invariant, as

$$T(Tv) = T^2v = -cTv - dv \in U.$$

If T– and T– were linearly dependent, then T– would be an eigenvector, contradicting the hypothesis; hence V–. The roots of the quadratic are V–, with V–. Using the relations for the sum and product of the roots,

$$c = -2a, \qquad V - -.$$

Thus,

$$T^2 - 2aT + (a^2 + b^2)V - -$$

on $V-$. We seek a basis of $V-$ in which

$$[T|_U] = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Let $V-$ and define

$$u_2 = \frac{1}{b}(T - aI)u_1.$$

By definition, $V-$. Now compute $V-$:

$$Tu_2 = T\left(\frac{1}{b}(T - aI)u_1\right) = \frac{1}{b}(T^2 - aT)u_1.$$

Substituting $v \in U_3-$,

$$Tu_2 = \frac{1}{b}\big(aTu_1 - (a^2 + b^2)u_1\big) = \frac{a}{b}(au_1 + bu_2) - \frac{a^2 + b^2}{b}u_1 = -bu_1 + au_2.$$

This matches the desired matrix representation. Geometrically, $v \in U_3-$ acts as a rotation combined with a scaling on the 2D plane $v \in U_3-$. Since it rotates every line without fixing any, it cannot have a real eigenvector; this is why a 2-dimensional invariant plane must appear.

3 Problem 3

(a) We are told

$$U = \text{span}(|v\rangle, T|v\rangle, T^2|v\rangle)$$

and

$$T^3|v\rangle = 3T^2|v\rangle - 2T|v\rangle - |v\rangle.$$

To show $v \in U_3-$ is $v \in U_3$—invariant, we simply check the images of the basis vectors under $v \in U_3-$. Clearly $v \in U_3-$ and $v \in U_3-$. Applying $v \in U_3-$ to the third basis vector yields

$$T(T^2|v\rangle) = T^3|v\rangle = -|v\rangle - 2T|v\rangle + 3T^2|v\rangle,$$

which is a linear combination of the basis vectors and thus lies in $v \in U_3-$. Hence $(T^2 + I)v = 0-$ is $(T^2 + I)v = 0$—invariant. **(b)** With the ordered basis $(T^2 + I)v = 0-$, the images under $(T^2 + I)v = 0-$ are:

$$\begin{aligned} T(|v\rangle) &= 0|v\rangle + 1T|v\rangle + 0T^2|v\rangle, \\ T(T|v\rangle) &= 0|v\rangle + 0T|v\rangle + 1T^2|v\rangle, \\ T(T^2|v\rangle) &= -1|v\rangle - 2T|v\rangle + 3T^2|v\rangle. \end{aligned}$$

Hence the matrix representation is

$$[T|_U]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & -2 \\ 0 & 1 & 3 \end{pmatrix}.$$

(c) The minimal polynomial is the monic polynomial of smallest degree annihilating $(T^2 + I)v = 0-$. The given equation implies

$$(T^3 - 3T^2 + 2T + I)|v\rangle = 0.$$

Since $(T^2+I)v = 0-$ are linearly independent, no polynomial of degree 2 or less can annihilate $(T^2+I)v = 0-$. Therefore the minimal polynomial is

$$p(x) = x^3 - 3x^2 + 2x + 1.$$

(d) The characteristic polynomial of $(T^2 + I)v = 0-$ equals its minimal polynomial (both have degree 3). Every cubic polynomial with real coefficients has at least one real root (by the Intermediate Value Theorem), so $(T^2 + I)v = 0-$ has at least one real eigenvalue, hence an eigenvector.

4 Problem 4

We are told $(T^2 + I)v = 0$– and $Tv \in U_3$– for $Tv \in U_3$–. **(a)** Apply Gram-Schmidt:

$$u_1 = v_1 \implies \|u_1\| = \sqrt{2}, \qquad |e_1\rangle = \frac{1}{\sqrt{2}}|v_1\rangle.$$

Next,

$$u_2 = v_2 - \langle e_1|v_2\rangle e_1, \qquad \langle e_1|v_2\rangle = \frac{1}{\sqrt{2}}(1).$$

So

$$u_2 = v_2 - \frac{1}{2}v_1.$$

Also,

$$\|u_2\|^2 = \langle v_2 - \tfrac{1}{2}v_1 \mid v_2 - \tfrac{1}{2}v_1\rangle = 2 - \frac{1}{2}(1) - \frac{1}{2}(1) + \frac{1}{4}(2) = \frac{3}{2}.$$

Thus

$$|e_2\rangle = \sqrt{\frac{2}{3}}\left(|v_2\rangle - \frac{1}{2}|v_1\rangle\right).$$

For the third vector,

$$u_3 = v_3 - \langle e_1|v_3\rangle e_1 - \langle e_2|v_3\rangle e_2.$$

Now

$$\langle e_1|v_3\rangle = \frac{1}{\sqrt{2}}, \qquad \langle e_2|v_3\rangle = \sqrt{\frac{2}{3}}\langle v_2 - \tfrac{1}{2}v_1 \mid v_3\rangle = \sqrt{\frac{2}{3}}\left(1 - \frac{1}{2}\right) = \frac{1}{\sqrt{6}}.$$

Thus

$$u_3 = v_3 - \frac{1}{2}v_1 - \frac{1}{6}(2v_2 - v_1) = v_3 - \frac{1}{3}v_2 - \frac{1}{3}v_1.$$

Finally,

$$\|u_3\|^2 = \frac{1}{9}\|3v_3 - v_2 - v_1\|^2 = \frac{1}{9}(18 + 2 + 2 - 6 - 6 + 2) = \frac{12}{9} = \frac{4}{3}.$$

So

$$|e_3\rangle = \frac{\sqrt{3}}{2}\left(|v_3\rangle - \frac{1}{3}|v_2\rangle - \frac{1}{3}|v_1\rangle\right).$$

(b) Expressing $Tv \in U_3$– in the $Tv \in U_3$– basis:

$$\begin{aligned}
|v_1\rangle &= \sqrt{2}|e_1\rangle, \\
|v_2\rangle &= \sqrt{\frac{3}{2}}|e_2\rangle + \frac{1}{\sqrt{2}}|e_1\rangle, \\
|v_3\rangle &= \frac{2}{\sqrt{3}}|e_3\rangle + \frac{1}{\sqrt{6}}|e_2\rangle + \frac{1}{\sqrt{2}}|e_1\rangle.
\end{aligned}$$

(c) Project

$$|v\rangle = |v_1\rangle + 2|v_2\rangle - |v_3\rangle$$

onto $Tv \in U_3$–, which is exactly $Tv \in U_3$–. First, rewrite $Tv \in U_3$– in the $Tv \in U_3$– basis:

$$\begin{aligned}
|v\rangle &= \left(\sqrt{2} + \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)|e_1\rangle + \left(\frac{2\sqrt{3}}{\sqrt{2}} - \frac{1}{\sqrt{6}}\right)|e_2\rangle - \frac{2}{\sqrt{3}}|e_3\rangle \\
&= \frac{3\sqrt{2}}{2}|e_1\rangle + \frac{5\sqrt{6}}{6}|e_2\rangle - \frac{2}{\sqrt{3}}|e_3\rangle.
\end{aligned}$$

The projection is

$$P(|v\rangle) = \frac{3\sqrt{2}}{2}|e_1\rangle + \frac{5\sqrt{6}}{6}|e_2\rangle.$$

In terms of the original vectors:

$$P(|v\rangle) = \frac{3}{2}|v_1\rangle + \frac{5}{3}\left(|v_2\rangle - \frac{1}{2}|v_1\rangle\right) = \frac{2}{3}|v_1\rangle + \frac{5}{3}|v_2\rangle.$$

(d) Let the projection be $Tv \in U_3$–. By orthogonality, the residual is perpendicular to $Tv \in U_3$– and $(T^2+I)(Tv) = 0$–, i.e.

$$\langle v_1|v\rangle = c_1\langle v_1|v_1\rangle + c_2\langle v_1|v_2\rangle \implies 3 = 2c_1 + c_2,$$
$$\langle v_2|v\rangle = c_1\langle v_2|v_1\rangle + c_2\langle v_2|v_2\rangle \implies 4 = c_1 + 2c_2.$$

Solving gives $(T^2+I)(Tv) = 0$–, $(T^2+I)(Tv) = 0$–. Thus

$$P(|v\rangle) = \frac{2}{3}|v_1\rangle + \frac{5}{3}|v_2\rangle.$$

5 Problem 5

We want to find

$$P_U = \alpha|u\rangle\langle u| + \beta|u\rangle\langle v| + \gamma|v\rangle\langle u| + \delta|v\rangle\langle v|.$$

Since $(T^2+I)(Tv) = 0$– projects onto $(T^2+I)(Tv) = 0$–, it must act as the identity on $(T^2+I)(Tv) = 0$–, i.e.

$$P_U|u\rangle = |u\rangle, \qquad P_U|v\rangle = |v\rangle.$$

Let

$$\langle u|u\rangle = 1, \qquad \langle v|v\rangle = 1, \qquad \langle u|v\rangle = c, \qquad |c| < 1.$$

Note that in a complex inner product space, $(T^2+I)(Tv) = 0$–, where $(T^2+I)(Tv) = 0$– denotes the complex conjugate of $(T^2+I)(Tv) = 0$–. We work with the general complex case first. Now

$$P_U|u\rangle = (\alpha + \beta c)|u\rangle + (\gamma + \delta\overline{c})|v\rangle = |u\rangle,$$

so

$$\alpha + \beta(T^2+I)(Tv) = 0 - -, \qquad \gamma + \delta\overline{c} = 0.$$

Similarly,

$$P_U|v\rangle = (\alpha c + \beta)|u\rangle + (\gamma c + \delta)|v\rangle = |v\rangle,$$

so

$$\alpha c + \beta = 0, \qquad \gamma c + \delta = 1.$$

From $p_1(x) = x - 1$–, we get $p_1(x) = x - 1$–. Substituting into $p_1(x) = x - 1$–:

$$\alpha(1 - |c|^2) = 1 \implies \alpha = \frac{1}{1-|c|^2}, \qquad \beta = -\frac{c}{1-|c|^2}.$$

From $p_1(x) = x - 1$–, we get $p_1(x) = x - 1$–. Substituting into $p_1(x) = x - 1$–:

$$\delta(1 - |c|^2) = 1 \implies \delta = \frac{1}{1-|c|^2}, \qquad \gamma = -\frac{\overline{c}}{1-|c|^2}.$$

Thus

$$\boxed{P_U = \frac{1}{1-|c|^2}\Big(|u\rangle\langle u| - c|u\rangle\langle v| - \overline{c}|v\rangle\langle u| + |v\rangle\langle v|\Big).}$$

If $p_1(x) = x - 1$– is real, then $p_1(x) = x - 1$–, and the coefficients reduce to

$$\alpha = \delta = \frac{1}{1-c^2}, \qquad \beta = \gamma = -\frac{c}{1-c^2}.$$

Verification.

$$P_U|u\rangle = \frac{1}{1-|c|^2}\Big(|u\rangle - c\overline{c}|u\rangle - \overline{c}|v\rangle + \overline{c}|v\rangle\Big) = |u\rangle.$$

$$P_U|v\rangle = \frac{1}{1-|c|^2}\Big(c|u\rangle - c|u\rangle - |c|^2|v\rangle + |v\rangle\Big) = |v\rangle.$$

For any $p_1(x) = x - 1-$, write $p_1(x) = x - 1-$ with $p_1(x) = x - 1-$. Since $p_1(x) = x - 1-$ and $p_1(x) = x - 1-$, applying the ket-bra operator directly gives $p_1(x) = x - 1-$. Thus

$$P_U(au + bv + w) = a(P_U u) + b(P_U v) + p_1(x) = x - 1 - -.$$

Applying $p_1(x) = x - 1-$ a second time:

$$P_U^2(x) = P_U(au + bv) = au + bv = P_U(x).$$

Hence $p_1(x) = x - 1-$.

6 Problem 6

(a) Any vector $p_1(x) = x - 1-$ can be uniquely written as

$$v = P_U|v\rangle + P_{U^\perp}|v\rangle.$$

Since we are given $p_1(x) = x - 1-$, the full description of all such vectors is

$$|v\rangle = |u\rangle + |w\rangle,$$

where $p_1(x) = x - 1-$ is any arbitrary vector in $p_1(x) = x - 1-$. **(b)** Let $p_1(x) = x - 1-$. By the Orthogonal Decomposition Theorem, we can write

$$|v\rangle = |x\rangle + |w\rangle,$$

where $p_1(x) = x - 1-$ and $p_1(x) = x - 1-$. Applying the projection gives $p_1(x) = x - 1-$. The given condition says $p_1(x) = x - 1-$, so $p_1(x) = x - 1-$. Replacing $p_1(x) = x - 1-$ in the initial decomposition proves that

$$|v\rangle = |u\rangle + |w\rangle, \qquad |w\rangle \in U^\perp.$$

(c) We are told $p_1(x) = x - 1-$ and $p_1(x) = x - 1-$. From (b),

$$|v\rangle = |e_1\rangle + |w\rangle$$

where $p_1(x) = x - 1-$. Since $p_1(x) = x - 1-$ is an orthonormal basis,

$$U^\perp = \text{span}(|e_3\rangle, |e_4\rangle).$$

Thus

$$|v\rangle = |e_1\rangle + a|e_3\rangle + b|e_4\rangle.$$

Applying the norm condition $p_1(x) = x - 1-$:

$$p_1(x) = x - 1 - - \implies p_1(x) = x - 1 - -.$$

Geometric description: the set of solutions forms a circle of radius $p_1(x) = x - 1-$ lying in the affine plane $p_1(x) = x - 1-$, i.e. the orthogonal plane $p_1(x) = x - 1-$ translated by the vector $p_1(x) = x - 1-$.

7 Problem 7

The vector $p_1(x) = x - 1$– minimizing $p_1(x) = x - 1$– is simply the orthogonal projection of $p_1(x) = x - 1$– onto $p_1(x) = x - 1$–. Let

$$|u\rangle = c_1|u_1\rangle + c_2|u_2\rangle.$$

Orthogonality requires

$$\langle u_1|v-u\rangle = 0, \qquad \langle u_2|v-u\rangle = 0.$$

Thus

$$\langle u_1|u\rangle = \langle u_1|v\rangle \implies c_1(2) + c_2(1) = 3,$$
$$\langle u_2|u\rangle = \langle u_2|v\rangle \implies c_1(1) + c_2(5) = 4.$$

Solving the system

$$2c_1 + c_2 = 3, \qquad c_1 + 5c_2 = 4,$$

we get $p_1(x) = x - 1$–, so

$$2(4 - 5c_2) + c_2 = 3 \implies 8 - 9c_2 = 3 \implies c_2 = \frac{5}{9}.$$

Hence

$$c_1 = 4 - 5\left(\frac{5}{9}\right) = \frac{11}{9}.$$

Therefore,

$$|u\rangle = \frac{11}{9}|u_1\rangle + \frac{5}{9}|u_2\rangle.$$

To find the minimum squared error without finding $p_1(x) = x - 1$–, use the Pythagorean theorem for orthogonal projections:

$$\||v\rangle - |u\rangle\|^2 = \|v\|^2 - \|u\|^2.$$

We compute $p_1(x) = x - 1$– as $p_1(x) = x - 1$–:

$$\|u\|^2 = \langle u|v\rangle = c_1\langle u_1|v\rangle + c_2\langle u_2|v\rangle = \frac{11}{9}(3) + \frac{5}{9}(4) = \frac{53}{9}.$$

Hence the minimum squared error is

$$\boxed{\|v\|^2 - \frac{53}{9}}.$$

Note: since $p_1(x) = x - 1$– is not specified in the problem, this is the best possible answer. If a numerical value is required, the value of $p_1(x) = x - 1$– must be given.

8 Problem 8

Prove $p_1(x) = x - 1$–**:** Let $p_1(x) = x - 1$–. By definition, for all $p_1(x) = x - 1$–, $p_1(x) = x - 1$–. Since $p_1(x) = x - 1$–, every $p_1(x) = x - 1$– is also in $p_1(x) = x - 1$–. Thus $p_1(x) = x - 1$– for all $p_1(x) = x - 1$–, so $p_1(x) = x - 1$–. **Prove** $p_1(x) = x - 1$–**:** Let $p_1(x) = x - 1$–. The projection $p_1(x) = x - 1$– produces a vector in $p_1(x) = x - 1$–. Since $p_1(x) = x - 1$–, this vector is also in $p_1(x) = x - 1$–. The projection of any vector residing in $p_1(x) = x - 1$– onto $p_1(x) = x - 1$– is the vector itself. Therefore,

$$P_W(P_U(x)) = P_U(x).$$

Prove $p_1(x) = x - 1$–**:** Let $p_1(x) = x - 1$–. By definition, $p_1(x) = x - 1$–. Applying $p_1(x) = x - 1$– to both sides gives

$$P_W(P_U(u)) = P_W(u).$$

By the hypothesis $p_1(x) = x - 1$–, the left-hand side evaluates to $p_1(x) = x - 1$–. Thus $p_1(x) = x - 1$–. Since the image of $p_1(x) = x - 1$– is exactly $p_1(x) = x - 1$–, we obtain $p_1(x) = x - 1$–. Hence $p_1(x) = x - 1$–.

Ket-bra interpretation: If we construct an ONB for $p_1(x) = x - 1$– such that the first $p_1(x) = x - 1$– vectors form an ONB for $p_1(x) = x - 1$– (i.e. $p_1(x) = x - 1$–), then

$$P_W = \sum |u_i\rangle\langle u_i| + \sum |w_j\rangle\langle w_j| = P_U + P_{W \ominus U}.$$

The operator product becomes

$$P_W P_U = (P_U + P_{W \ominus U})P_U = P_U^2 + 0 = P_U.$$

9 Problem 9

(a) Any orthonormal list of vectors in a finite-dimensional inner product space can be extended to an ONB of the entire space $p_1(x) = x - 1$–. Let the extended ONB be

$$\{|e_1\rangle, \ldots, |e_m\rangle, |e_{m+1}\rangle, \ldots, |e_n\rangle\},$$

where the latter $p_1(x) = x - 1$– vectors form a basis for $p_1(x) = x - 1$–. **(b)** The matrix representing $p_1(x) = x - 1$– is constructed by finding the coefficients $p_1(x) = x - 1$–. For $p_1(x) = x - 1$–, since $p_1(x) = x - 1$– is $p_1(x) = x - 1$—invariant, $p_1(x) = x - 1$– remains strictly in $p_1(x) = x - 1$–. Thus, for any $p_1(x) = x - 1$–, the coefficient $p_1(x) = x - 1$–. This forces the entire bottom-left block of the matrix (representing transitions from $p_1(x) = x - 1$– to $p_1(x) = x - 1$–) to be a zero matrix. Since $p_1(x) = x - 1$– is upper triangular, the top-left block $p_1(x) = x - 1$– is upper triangular, yielding the block structure

$$\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

(c) The orthogonal projections are $p_1(x) = x - 1$– and $p_1(x) = x - 1$–. The zero block in the matrix specifically represents the operator $p_2(x) = x - 2$–. The fact that $p_2(x) = x - 2$– strictly means that if a vector starts in $p_2(x) = x - 2$– (applied by $p_2(x) = x - 2$–), and is transformed by $p_2(x) = x - 2$–, it has exactly zero component in the orthogonal complement $p_2(x) = x - 2$– (applied by $p_2(x) = x - 2$–). This is the direct structural definition of $p_2(x) = x - 2$– being a $p_2(x) = x - 2$—invariant subspace.

10 Problem 10

(a) The vector space is the set of real-valued random variables with finite second moments ($p_2(x) = x - 2$– space). The subspace onto which $p_2(x) = x - 2$– is being projected is

$$\text{span}(1, X).$$

(b) By the Orthogonal Projection Theorem, the optimal residual $p_2(x) = x - 2$– minimizing the distance metric must be orthogonal to every vector in the subspace. Thus the residual is orthogonal to the basis vector 1, giving

$$\langle Y - \hat{Y}, 1\rangle = 0 \implies Y - \hat{Y} \perp 1.$$

It is also orthogonal to the basis vector $p_2(x) = x - 2$–, giving

$$\langle Y - \hat{Y}, X\rangle = 0 \implies Y - \hat{Y} \perp X.$$

(c) Expanding the orthogonality with 1:

$$\mathbb{E}[Y - (a + bX)] = 0 \implies \mathbb{E}[Y] - a - b\mathbb{E}[X] = 0 \implies a = \mathbb{E}[Y] - b\mathbb{E}[X].$$

Expanding the orthogonality with $p_2(x) = x - 2$–:

$$\mathbb{E}[(Y - a - bX)X] = 0 \implies \mathbb{E}[XY] - a\mathbb{E}[X] - b\mathbb{E}[X^2] = 0.$$

Substitute $p_2(x) = x - 2$–:

$$\mathbb{E}[XY] - (\mathbb{E}[Y] - b\mathbb{E}[X])\mathbb{E}[X] - b\mathbb{E}[X^2] = 0$$
$$\implies \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = b(\mathbb{E}[X^2] - \mathbb{E}[X]^2).$$

This simplifies directly to

$$\mathrm{Cov}(X, Y) = b\,\mathrm{Var}(X) \implies b = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}.$$

(d) We are told $p_2(x) = x - 2$– and $p_2(x) = x - 2$–. Then

$$\hat{Y} = \mathbb{E}[Y] - b\mathbb{E}[X] + bX = \mathbb{E}[Y] + bX_0.$$

Thus $p_2(x) = x - 2$–. Notice

$$b = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} = \frac{\mathbb{E}[X_0 Y_0]}{\mathbb{E}[X_0^2]} = \frac{\langle Y_0, X_0\rangle}{\langle X_0, X_0\rangle}.$$

Thus

$$\hat{Y} - \mathbb{E}[Y] = \frac{\langle Y_0, X_0\rangle}{\|X_0\|^2} X_0,$$

which is the geometric formula for the orthogonal projection of $p_2(x) = x - 2$– onto the 1-dimensional space generated by $p_2(x) = x - 2$–. **(e)** The updated predictor $p_2(x) = x - 2$– represents an orthogonal projection onto $p_2(x) = x - 2$–. Adding $p_2(x) = x - 2$– strictly decreases the minimum mean-square error iff the projection of $p_2(x) = x - 2$– onto this larger subspace differs from its projection onto $p_2(x) = x - 2$–. This strictly happens iff $p_2(x) = x - 2$– is not orthogonal to the component of $p_2(x) = x - 2$– orthogonal to $p_2(x) = x - 2$– (i.e. iff $p_2(x) = x - 2$– provides novel explanatory variance not captured by $p_2(x) = x - 2$–).

Problem 11

We must verify the inner product axioms for

$$\langle p, q\rangle = p(-1)q(-1) + p(0)q(0) + p(1)q(1) + p'(0)q'(0)$$

on $p_2(x) = x - 2$–.

1. **Symmetry:** Real number multiplication is commutative, thus

$$\langle p, q\rangle = \langle q, p\rangle.$$

2. **Linearity:**

$$\begin{aligned}\langle ap + br, q\rangle &= (ap + br)(-1)q(-1) + \cdots \\ &= a\big(p(-1)q(-1) + \cdots\big) + b\big(r(-1)q(-1) + \cdots\big) \\ &= a\langle p, q\rangle + b\langle r, q\rangle.\end{aligned}$$

3. **Positive Definiteness:**

$$\langle p, p\rangle = p(-1)^2 + p(0)^2 + p(1)^2 + p'(0)^2 \geq 0.$$

4. **Definiteness (Zero condition):** If $p_2(x) = x - 2$–, then each squared term must be zero. This forces

$$p(-1) = 0, \qquad p(0) = 0, \qquad p(1) = 0.$$

   Since $p_2(x) = x - 2$–, it is a polynomial of degree at most 2. A polynomial of degree $p_2(x) = x - 2$– possessing 3 distinct roots must strictly be the zero polynomial ($p_2(x) = x - 2$–).

Since all axioms are satisfied, $p_2(x) = x - 2$– is indeed a valid inner product on $p_2(x) = x - 2$–.